MUKESH K

Data Scientist | AI/ML Engineer

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PROFESSIONAL SUMMARY

B.Sc. Data Science graduate building LLM applications: retrieval-augmented generation, LLM-based extraction pipelines, and tool-calling agents with LangGraph and MCP. Evaluation-first: measure results, trace failures to root causes (invalid terminology codes, a wrong-patient merge that passed validation, a misread evaluation metric) and document what still doesn't work. Currently a Frontend Developer Intern at Conceptra Labs, working in React and Angular.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript, HTML/CSS
LLM / AI Engineering: Claude API, RAG, LangGraph, MCP (Model Context Protocol), LLM-as-a-Judge evaluation (DeepEval G-Eval), FAISS, Prompt-based Entity Extraction, Pydantic
Healthcare Data: FHIR R4B, Synthea, LOINC / RxNorm terminology mapping
Machine Learning: Scikit-learn, XGBoost, LightGBM, TF-IDF, Feature Engineering, Cross-Validation
Deep Learning / CV: PyTorch, Ultralytics YOLO11, YOLOv8
Web Development: FastAPI, REST APIs, SQLAlchemy, JWT Auth, React, Angular
Data & Databases: Pandas, NumPy, Matplotlib, Seaborn, SQLite, PostgreSQL, MySQL
Tools & Platforms: Git, GitHub, Claude Code, Jupyter, Google Colab, Kaggle Notebooks, Render

EXPERIENCE

Frontend Developer Intern — Conceptra Labs Aug 2026 – Present

AI/LLM & Agentic AI company

- Completed a structured frontend training plan (HTML, CSS, JavaScript, React, Angular) with daily reports, GitHub commits and weekly evaluation; moving onto client projects spanning frontend and backend from Nov 2026.
- Built client homepage UIs for a retail store and a spice-shop brand (Angular), and a React task manager using components, props, useState, lifted state, search and filters.

PROJECTS

Canonical Medical Record FHIR Pipeline 2026

Healthcare Data Engineering & LLM-Based Entity Extraction — internship assessment, LezDo TechMed

- Built a 7-stage pipeline (page classification, content extraction, Claude API entity extraction, terminology normalization, FHIR construction/validation, SQLite persistence, provenance) turning multi-document synthetic medical records into FHIR R4B resources; the final live run validated **243/243 resources** across 9 types and recovered **41/41** labeled test facts.
- Audited past structural validation: found **35 invalid codes** in the terminology dictionary (e.g. methocarbamol mapped to metolazone) and corrected all against NLM services; caught another patient's page merged into the record and fixed it with a majority-vote patient matcher with DOB-conflict detection.
- Replaced a hardcoded confidence score with per-fact extraction and mapping confidence, routing low-confidence facts to a review queue; every resource traces to its source page and snippet; remaining limits documented in FAILURES.md.

Tech: Python | Claude API | FHIR R4B | Synthea | SQLite | Pydantic | **Link:** github.com/Mukesh-2005/project-3-fhir-pipeline

Multi-Class Vehicle Detection System (YOLO11l) Jun 2026

Computer Vision & Object Detection

- Trained YOLO11l to detect 5 vehicle classes common to Indian traffic (car, bus, truck, motorcycle, auto-rickshaw) on 6,928 images, reaching a validated **mAP50 of 0.459**; moved training from Colab to Kaggle after repeated disconnections and GPU quota limits.
- Corrected a 20:1 car-class dominance for the YOLOv8s baseline by cutting a 40K-image, 20-class pool to a balanced ~1,300 images, a deliberate size-for-balance tradeoff under free-tier GPUs.
- Documented the main failure mode (lower precision when cars occlude motorcycles and auto-rickshaws in dense traffic) and fixed an evaluation script reading per-class mAP50 from the wrong attribute.

Tech: YOLO11 | YOLOv8 | PyTorch | Ultralytics | Kaggle | **Link:** github.com/Mukesh-2005/traffic-vehicle-detection

EDUCATION

B.Sc. Data Science — Manonmaniam Sundaranar University, Tirunelveli 2023 – 2026

CERTIFICATIONS

IBM Data Science Professional Certificate (Specialization), 2026 • Machine Learning with Python — IBM, 2024 • Databases and SQL for Data Science — IBM, 2024 • Python for Data Science, AI & Development — IBM, 2025